

QUESTION 1

What is the value of $267_8 + 104_8$?

- A. 373_8 B. 374_8 C. 371_8 D. $36A_8$ E. 163_8

QUESTION 2

What is the value of `z` after executing the code to the right?

- A. 10 B. 300 C. 1464
D. 70 E. 264

```
int x = 10;
int y = x * 4 + 6;
int z = y + 4 * 6;
```

QUESTION 3

What are the values of `ch` and `total` after executing the code to the right?

- A. `ch: 'a'` B. `ch: 'b'`
 `total: 25` `total: 24`
C. `ch: 'a'` D. `ch: 'b'`
 `total: 26` `total: 25`
E. `ch: 'c'`
 `total: 24`

```
int total = 0;
char ch = 'z';

while (ch >= 'b') {
    ++total;
    --ch;
}
```

QUESTION 4

Which of these can be used inside any class to output data member `x` from `MyClass m`?

- A. `System.out.print(m(x))`
B. `System.out.print(m.getX())`
C. `System.out.print(m/x)`
D. `System.out.print(m.x)`
E. Both B and D

```
public class MyClass {
    public MyClass(double xInit, int yInit) {
        x = xInit;
        y = yInit;
    }

    public double getX() { return x; }
    public int getY() { return y; }

    private double x;
    public int y;
}
```

QUESTION 5

Which of these builds a `MyClass` object with data member `x` initialized to 0.7 and data member `y` initialized to 12?

- A. `MyClass m = (0.7, 12)`
B. `MyClass m = {0.7, 12}`
C. `MyClass m(0.7, 12)`
D. `MyClass(0.7, 12)`
E. `new MyClass(0.7, 12)`

QUESTION 6

Which of these subtracts 1 from `int x` and returns the previous value?

- A. `x-1?x` B. `x-1` C. `x--` D. `--x` E. Both A and D

QUESTION 7 What is output by the code to the right? A. e B. a C. eta D. t E. nothing	<pre>String s = "alpha-beta"; System.out.print(s.charAt(8));</pre>
QUESTION 8 What replaces <*1> in the code to the right to give the character at index <i>i</i> in String <i>s</i>? A. <i>s+i</i> B. <i>s[i]</i> C. <i>s.index(i)</i> D. <i>s.charAt(i)</i> E. Both B and D	<pre>public static int compute(String s) { int total = 0; for (int i=0; i < s.length(); ++i) { char c = <*1>; if (c >= '0' && c <= '9') total = total + (c - '0'); } return total; }</pre>
QUESTION 9 Assume <*1> is filled in correctly. What is returned by <code>compute("1999")</code>? A. 28 B. 1 C. 9 D. 729 E. 2006	
QUESTION 10 What is the value of <code>total</code> after executing the code to the right? A. -2 B. -1 C. 0 D. 1 E. 2	<pre>int x = 42, total = 0; while (x != 0) { switch(x%3) { case 0: total = total + 1; break; case 2: total = total - 1; break; } x = x/3; }</pre>
QUESTION 11 What is returned by <code>process(3)</code>? A. "abaaba" B. "ababab" C. "aaaaaa" D. "babbab" E. "bbb bbb"	<pre>public static String process(int x) { if (x == 0) return ""; else if (x<0) return "a" + process(-x-1) + "a"; else return "b" + process(-x+1) + "b"; }</pre>
QUESTION 12 How many a's are in the string returned by <code>process(-1001)</code>? A. 500 B. 501 C. 1000 D. 1001 E. 1002	

QUESTION 13

Can these static methods all be part of the same class?

- A. Yes
- B. No, because the names of the methods are the same
- C. No, because the first and second method have the same name, return type, and parameter names
- D. No, because the methods have the same name and they all have two parameters
- E. No, because the first and third method have the same name and parameter types

```
public static int f(int x, int y) {
    // code not shown
}

public static int f(int x, double y) {
    // code not shown
}

public static double f(Integer x,
                        Integer y) {
    // code not shown
}
```

QUESTION 14

Which of these replaces **<*1>** in the code to the right to create arrays that have only one copy for the class, not one for each instance, and that are accessible only inside the Date class?

- A. `int[12]`
- B. `private int[]`
- C. `private static int[]`
- D. `static int[12]`
- E. More than one of these

```
public class Date {
    public Date(int day, int month, int year)
    {
        if (year <= 0)
            throw new IllegalArgumentException(
                "Bad year");
        this.year = year;

        if (month <= 0 || month > 12)
            throw new IllegalArgumentException(
                "Bad month");

        if (isLeapYear(year)) {
            this.dayofyear =
                leapoffset[month-1]+day;
            if (dayofyear>leapoffset[month])
                throw new IllegalArgumentException(
                    "Bad day");
        }
        else {
            this.dayofyear = offset[month-1]+day;
            if (dayofyear>offset[month])
                throw new IllegalArgumentException(
                    "Bad day");
        }
    }
}
```

QUESTION 15

Assume **<*1>** is filled in correctly. Which of these causes an `IllegalArgumentException` to be thrown?

- A. `new Date(31, 7, 2000)`
- B. `new Date(31, 7, 2100)`
- C. `new Date(29, 2, 2000)`
- D. `new Date(29, 2, 2100)`
- E. More than one of these

```
private static boolean isLeapYear(int y)
{
    if (y%400 == 0) return true;
    if (y%100 == 0) return false;
    if (y%4 == 0) return true;
    return false;
}

// other methods not shown

private int dayofyear, year;
<*1> offset = {0, 31, 59, 90, 120, 151, 181,
              212, 243, 273, 304, 334, 365};
<*1> leapoffset = {0, 31, 60, 91, 121, 152, 182,
                  213, 244, 274, 305, 335, 366};
}
```

QUESTION 16

What replaces **<*1>** in the code to the right to remove the first element from `elements` and return its value?

- A. `elements.removeFirst(); return;`
- B. `return elements.removeFirst();`
- C. `return elements.remove(length);`
- D. `return elements.remove(1);`
- E. `return elements.dequeue();`

For the remaining questions, assume that **<*1>** has been filled in correctly.

QUESTION 17

What is output by static method `test()`?

- A. 312 B. 321 C. 231
- D. 123 E. 132

QUESTION 18

What is the total running time of using the `enqueue()` method to insert n items? Choose the most restrictive correct answer.

- A. $O(1)$ B. $O(\log n)$ C. $O(n)$
- D. $O(n^2)$ E. $O(n \log n)$

QUESTION 19

On which of these arrays would static method `process()` return 3 if the array was passed as its parameter?

- A.

1	2	3	4	5	6
---	---	---	---	---	---
- B.

6	5	4	3	2	1
---	---	---	---	---	---
- C.

1	3	5	2	4	6
---	---	---	---	---	---
- D.

2	4	6	1	3	5
---	---	---	---	---	---
- E.

1	3	5	6	4	2
---	---	---	---	---	---

```
public class MyQueue<E> {

    public MyQueue() {
        elements = new LinkedList<E>();
    }

    public void enqueue(E element) {
        elements.addLast(element);
    }

    public E dequeue() {
        <*1>
    }

    public boolean isEmpty() {
        return elements.size() == 0;
    }

    private LinkedList<E> elements;

    public static void test() {
        MyQueue<Integer> testQ =
            new MyQueue<Integer>();
        testQ.enqueue(1);
        testQ.enqueue(2);
        testQ.enqueue(3);
        System.out.print(testQ.dequeue());
        System.out.print(testQ.dequeue());
        System.out.print(testQ.dequeue());
    }

}
```

```
public static int process(int[] array) {
    int returnValue = 0;
    for (int i=array.length; i>0; --i)
        if (array[i]>array[i-1])
            ++returnValue;
    return returnValue;
}
```

QUESTION 20

What replaces <*1> and <*2> in the code to the right so that it is syntactically correct?

- A. <*1>: abstract class
<*2>: extends
- B. <*1>: interface
<*2>: extends
- C. <*1>: abstract class
<*2>: implements
- D. <*1>: interface
<*2>: implements
- E. Either A or D

```
public <*1> Shape {
    public abstract double area();
    public abstract double perimeter();
}

public class Circle <*2> Shape {
    public Circle(double radius) {
        r = radius;
    }
    public double area() {
        return Math.PI * r * r;
    }
    public double perimeter() {
        return 2 * Math.PI * r;
    }
    private double r;
}
```

QUESTION 21

Assume <*1> and <*2> are filled in correctly. Which of these can be used in any class to output the area of Circle c?

- A. System.out.print(c.area())
- B. System.out.print(area(c))
- C. System.out.print(c.area(r))
- D. System.out.print(Math.PI*c.r*c.r)
- E. All of these

QUESTION 22

What is output by the code to the right?

- A. 123458 B. 1234589
- C. 123459 D. 12345589
- E. The code contains an infinite loop

```
int [] intArray = {1, 2, 3, 4, 5};
int i = 0;
try {
    while (true) {
        System.out.print(intArray[i]);
        ++i;
    }
} catch (Exception e) {
    System.out.print(8);
} finally {
    System.out.print(9);
}
```

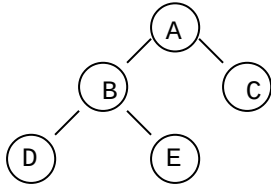
QUESTION 23

What is output by the code to the right?

- A. 89 B. 19
- C. 98 D. 99
- E. 9

```
int x = 3^2;
int y = (int)Math.pow(3.0,2.0);

System.out.print("" + x + y);
```

QUESTION 24 Which of these input strings would cause the code to the right to set x to 12, y to 0.2, and s to " 19"? A. 120.219 B. 12 0.2 19 C. 12.2 19 D. 12.219 E. Both B and C	<pre>Scanner input = new Scanner(System.in); int x = input.nextInt(); double y = input.nextDouble(); String s = input.nextLine();</pre>
QUESTION 25 What is output by the code to the right? A. catfalse B. cattrue C. catchfalse D. cat0 E. An exception is thrown	<pre>String s = "cat", t = "catch"; System.out.print(t.substring(0,3)); System.out.print(s.equals(t.substring(0,3)));</pre>
QUESTION 26 What replaces <*1> to call the other MyClass constructor with the parameter 0? A. 0; B. MyClass(0); C. this(0); D. super(0); E. MySuperClass(0);	<pre>public class MyClass extends MySuperClass { public MyClass() { <*1> } public MyClass(int i) { // code not shown } // other data and methods not shown }</pre>
QUESTION 27 What order are the nodes processed in an in-order traversal of this binary tree? A. A B C D E B. E D C B A C. D B E A C D. E D A C B E. A B D E C	 <pre> graph TD A((A)) --- B((B)) A --- C((C)) B --- D((D)) B --- E((E)) </pre>
QUESTION 28 What is returned by m.keySet()? A. A Set containing Map.Entry objects B. A Set containing Integer objects C. A Set containing String objects D. An Iterator E. A Map from Integer objects to String objects	<pre>Map<Integer, String> m = new TreeMap<Integer, String>();</pre>

QUESTION 29

Which of these best describes the static method `sort()` to the right?

- A. Sorts in ascending order by finding the smallest remaining item each pass of the outer loop
- B. Sorts in ascending order by finding the largest remaining item each pass of the outer loop
- C. Sorts in descending order by finding the smallest remaining item each pass of the outer loop
- D. Sorts in descending order by finding the largest remaining item each pass of the outer loop
- E. Does not modify the array

```
public static void sort(Comparable[] array)
{
    for (int i=0; i<array.length-1; ++i) {
        Comparable c = array[i];
        int cIndex = i;
        for (int j=i+1; j<array.length; ++j)
            if (c.compareTo(array[j])>0) {
                c = array[j];
                cIndex = j;
            }
        array[cIndex] = array[i];
        array[i] = c;
    }
}
```

QUESTION 30

What is the running time of `sort()` on an array containing n items? Choose the most restrictive correct answer.

- A. $O(1)$
- B. $O(\log n)$
- C. $O(n)$
- D. $O(n^2)$
- E. $O(n \log n)$

QUESTION 31

What replaces `<*1>` in the code to the right to cause all of the items in `intArray` to be output?

- A. `for (iterator<Integer> item in data)`
- B. `for (item : data)`
- C. `for (Integer item : data)`
- D. `for (Integer item :
 ArrayList<Integer> data)`
- E. `for (Integer item in data)`

```
ArrayList<Integer> data;
// code to initialize data not shown

<*1> {
    System.out.println(item);
}
```

QUESTION 32

What replaces `<*1>` in the code to the right to check whether the character at position `i` in `s` is upper case?

- A. `Character.toUpperCase(s.charAt(i))`
- B. `Character.toUpperCase(s.charAt(i))`
- C. `Character.isUpperCase(s.charAt(i))`
- D. `s.charAt(i) + 'A' - 'a'`
- E. `s.charAt(i) + isLowerCase?32:0`

```
public static int numberUpper(String s) {
    int i = 0, total = 0;
    while (i < s.length()) {
        if (<*1>) ++total;
        ++i;
    }
    return total;
}
```

QUESTION 33

What replaces `<*1>` and `<*2>` in the code to the right so that all of `s` up to but not including the last space character is put in `firstName` and all of `s` after but not including the last space character is put in `lastName`? (Assume the loop to set `index` finds a space character.)

- A. `<*1>`: `s.substring(0, index-1)`
`<*2>`: `s.substring(index)`
- B. `<*1>`: `s.substring(0, index)`
`<*2>`: `s.substring(index-1)`
- C. `<*1>`: `s.substring(0, index)`
`<*2>`: `s.substring(index)`
- D. `<*1>`: `s.substring(0, index)`
`<*2>`: `s.substring(index+1)`
- E. `<*1>`: `s.substring(0, index-1)`
`<*2>`: `s.substring(index+1)`

```
public class Employee {

    public Employee(String s, int id) {
        idNumber = id;
        int index == s.length()-1;
        while (s.charAt(index) != ' ') --index;
        firstName = <*1>;
        lastName = <*2>;
    }

    public String toString() {
        return <*3>;
    }

    private String firstName, lastName;
    private int idNumber;
}
```

QUESTION 34

Assume `<*1>` and `<*2>` are filled in correctly. What replaces `<*3>` in the code to the right to give a string containing an employee's last name, a comma, the first name, and the ID number in brackets?

- A. `lastName + "," + firstName + "[" + idNumber + "]"`
- B. `lastName + ',' + firstName + '[' + (String)idNumber + '']'`
- C. `"lastName, firstName[idNumber]"`
- D. `lastName, firstName[idNumber]`
- E. `lastName + "," + firstName + "[" + (String)idNumber + "]"`

QUESTION 35

Which of these characters in a regular expression indicates that the previous group should appear zero or more times?

- A. -
- B. /
- C. .
- D. *
- E. +

QUESTION 36

How many `*`'s are output by the code to the right?

- A. 0
- B. 4
- C. 5
- D. 19
- E. 20

```
int x = 10;
do {
    System.out.print('*');
    if (--x == 7) break;
    System.out.print('*');
} while (x > 0);
```

QUESTION 37

What replaces `<*1>` in the code to the right to return true if the ratio of square feet to number of rooms is smaller than min? Use floating point arithmetic for the division.

- A. `((double)squareFeet)/numRooms < min`
- B. `(double)(squareFeet/numRooms) < min`
- C. `(double)(squareFeet)/numRooms < min`
- D. `double(squareFeet)/numRooms < min`
- E. Both A and C

```
public class Deed {
    // constructor and some methods not shown

    public boolean smallRooms(double min) {
        return <*1>;
    }

    private String address;
    private int squareFeet;
    private int numRooms;
}
```

QUESTION 38

Suppose `Deed` does not override the `toString()` method. In which class is the `toString()` method located that is called when a `Deed` object is output with `System.out.print()`?

- A. `Object`
- B. `System`
- C. `Deed`
- D. `IO`
- E. Printing is an invalid operation on `Deed` objects

QUESTION 39

On which of these arrays does the static method call `search(array, 6)` return 4?

- A.

1	3	5	5	8	12
---	---	---	---	---	----
- B.

2	3	4	5	6	7
---	---	---	---	---	---
- C.

18	3	24	12	6	10
----	---	----	----	---	----
- D. Both A and B
- E. Both B and C

```
public static int search(int[] array,
                        int item) {
    int index = 0;
    while (array[index++] < item);
    return index;
}
```

QUESTION 40

Which of these is the binary equivalent of the hexadecimal number 19_{16} ?

- A. 10001_2
- B. 01101_2
- C. 10101_2
- D. 10011_2
- E. 11001_2

Computer Science Answer Key

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1. A	11. D	21. A	31. C
2. D	12. E	22. B	32. C
3. A	13. A	23. B	33. D
4. E	14. C	24. B	34. A
5. E	15. D	25. B	35. D
6. C	16. B	26. C	36. C
7. D	17. D	27. C	37. E
8. D	18. C	28. B	38. A
9. A	19. E	29. A	39. D
10. C	20. E	30. D	40. E